

Digital finance legislation: Overview and state of play

SUMMARY

Digital finance can broadly be defined as financial services and instruments that use or are based on new information and communication technologies (ICT). A wide range of segments of the financial system are therefore concerned, from digital payment services to the new market infrastructures of crypto-assets using distributed ledger technologies (DLT). Policymakers expect digital finance to benefit the financial system – for example, in terms of transaction and settlement costs, as well as financial inclusion. However, digital finance also poses new risks, especially for financial stability and the protection of citizens.

The idea behind the EU regulatory approach is that by providing a sound regulatory framework, homogenous throughout the EU, these risks can be monitored and controlled, while also favouring the desired innovation. To that end, the European Commission, together with the European supervisory authorities and the European Central Bank, conduct regular reviews of the EU regulatory framework and check its ability to face these risks and the potential needs for intervention. In 2020, the Commission tabled a major digital finance strategy to provide a sound, EU-level regulatory and supervisory framework in a number of digital finance domains.

The EU has already adopted new laws resulting from this initiative. The Regulation on Markets in Crypto-assets is establishing a new legal environment for DLT-based 'coins' with a stable value ('stablecoins'); another regulation will provide a framework for the monitoring and control of digital operational resilience for the financial sector. More legislative procedures are ongoing in the fields of open finance and the digital euro. New directions are being suggested, such as the establishment of a 'unified ledger', to smoothen transfers between instruments using different DLT market infrastructure.



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Introduction

'[Digital finance](#)' is a term designating financial products and instruments, which use or are based on new technologies – broadly understood to be information and communication technologies (**ICT**). Therefore, digital finance is chiefly identified by the use of 'innovative' ICTs, which support 'traditional' financial services and instruments; digital innovation may also allow the development of new instruments like 'crypto-assets'. Digital finance is evolving rapidly, and innovations may have an impact on diverse areas of the financial system, from payment services to the financial markets. The most notable innovation is probably the introduction of decentralised distributed ledger technologies (**DLT**) and the fast-growing markets of DLT-based assets such as bitcoins and other crypto-assets.

Digital finance brings significant benefits to the financial system in various domains, including lower transaction costs and a higher level of financial inclusion. However, innovative digital finance tools may also pose new risks, especially regarding the stability of the financial system and the protection of citizens, and identifying these risks is sometimes a challenge for policymakers.¹ Finally, the fragmented regulatory approach not only creates complexity for EU market players but also has potential consequences for the level playing field; innovative digital finance thus requires EU-level legislation.²

Existing EU law already regulates some of the new digital financial instruments. For instance, the EU Regulation on Markets in Financial Instruments ([MiFIR](#)) and the Second Directive on Markets in Financial Instruments ([MiFID II](#)) cover the financial instruments based on DLT; however, MiFID II was recently amended to ensure that financial instruments 'issued by means of DLT' are indeed within the scope of the definition of financial instruments.³

2020 digital finance package

The 2020 [digital finance package](#) is the first regulatory package in this area tabled by the European Commission to complement the existing legislation. It builds upon the [digital finance strategy \(DFS\)](#) adopted in September 2020. The DFS defines the overarching strategic objective of the Commission of embracing digital finance, and sets out [four priorities](#):

- tackling fragmentation in the digital single market for financial services;
- ensuring that the EU regulatory framework facilitates digital innovation in the interest of consumers and market efficiency;
- creating an EU financial data space to promote data-driven innovation;
- addressing new challenges and risks associated with the digital transformation.

The DFS explicitly envisages that digital technologies will be key for modernising the EU economy across sectors. The objective is thus to foster the development of digital finance within a legal framework, which gives confidence to investors, consumers and businesses. To this end, users of financial services need to be protected against the risks stemming from expanding digital finance; the DFS therefore aims to foster innovation and protect users. The Commission has based its resulting digital finance package around four pillars, namely:

- (1) resilience of financial institutions;
- (2) markets in crypto-assets;
- (3) a digital euro;
- (4) payment services.

Digital Operational Resilience Act (DORA)

The [Regulation on digital operational resilience for the financial sector](#) – also known as the Digital Operational Resilience Act ([DORA](#)) – entered into force in January 2023. DORA fills the gaps in EU regulation as regards operational risks relating to ICT. It thus introduces and harmonises key digital operational requirements to make ICT operations resilient against severe operational disruption and cyber-attacks. DORA requires financial entities above a certain size to have in place governance and control measures that ensure prudent management of ICT risks. It also sets up control bodies and procedures for the arrangements. DORA requires detailed identification and documentation of ICT risks, and their management, and provides for testing as well as third-party risks.

The regulation covers a wide range of regulated financial entities, including credit institutions, payment and electronic money institutions, crypto-asset service providers, central securities depositories, trading venues and trade repositories. While DORA offers a general framework, the relevant European supervisory authorities (**ESAs**) are developing the technical standards; supervision and enforcement remain the competence of the national competent authorities (**NCA**s). The law will apply from 17 January 2025.

Crypto-assets

On 22 June 2022, the EU adopted the [Regulation on a pilot regime for market infrastructures based on DLT](#), which aims to [establish](#) a temporary common EU pilot regime for DLT-based financial services.⁴ The DLT pilot targets all crypto-assets, especially transferable securities, and would remove regulatory barriers to the issuing, trading and settlement of crypto-assets to help regulators gain experience in the use of DLT.

The [regulation](#) covers the granting, withdrawal and modification of permission – including exemptions and compensatory or corrective measures – to operate DLT market infrastructure, as well as the operation and supervision of DLT market infrastructure. It also provides for cooperation between DLT market infrastructure operators, national authorities and the ESMA.⁵

Regulatory sandbox

The UK Financial Conduct Authority (FCA) [sandbox](#) is considered a landmark for regulatory sandboxes; it was launched in 2016 and is still active. The EU [Blockchain Regulatory Sandbox](#) for innovative use cases involving DLT establishes a pan-EU framework for regulatory dialogues; blockchain is one type of DLT. It was [launched](#) in February 2023 and aims to facilitate the dialogue between regulators and innovators. Legal advice and regulatory guidance is provided in a safe and confidential environment. The dialogues are held across industry sectors and geographic regions, which help to identify and communicate best practices. The sandbox accepts cohorts of 20 blockchain use cases annually. Use cases are selected according to the maturity of the business case, legal and regulatory relevance, and their contribution to the policy priorities.

The [Regulation on Markets in Crypto-assets \(MiCA\)](#) entered into force in June 2023 and addresses regulatory gaps regarding DLT-based financial instruments. MiCA focuses almost exclusively on crypto-assets whose value is 'stabilised' against one or more assets, in particular currencies. MiCA identifies three categories of crypto-assets according to the right they represent.

- 'e-money tokens' (**EMT**) are crypto-assets whose value is 'stabilised' by referencing only one official currency;
- asset-referenced tokens (**ART**) are crypto-assets whose value is stabilised by referencing another value or right, including one or several official currencies;
- the third type covers all crypto-assets other than EMT and ART.

By enhancing the protection of consumers and investors as well as financial stability, the regulation is expected to promote innovation and the use of crypto-assets. MiCA [regulates](#) issuance and

trading of crypto-assets as well as the management of the underlying assets, where applicable, with additional regulatory rules aimed at 'significant' ART and EMT.

MiCA includes a number of Level 2 measures that must be developed before the regulation enters into force – by 30 December 2024, at the latest.⁶

Open finance

The European Commission tabled a package of proposals for [financial data access and payments](#)⁷ in June 2023. The package comprises three proposals touching upon data access, on the one hand, and payments services, on the other.

The first [proposal](#) on financial data access seeks to establish a framework for accessing individual and business customer data across a wide range of financial services. The ultimate objective is to support financial innovation and competition. To this end, the framework aims to ensure that all consumers and firms have effective instruments of control over their financial data. It also provides additional tools to ensure that personal data protection is aligned with the [General Data Protection Regulation](#). The framework would provide rights and obligations, including:

- (a) the option for customers to share their data with data users;
- (b) the obligation for customer data holders to make these data available to data users;
- (c) full control by customers over who accesses their data and for what purpose;
- (d) standardisation of customer data and the required technical interfaces.

The package is also composed of two proposals concerning payments, the first of which would amend the [payment services regime](#). The second [proposal](#) would establish a payment services regulation. These proposals are meant to ensure continuity in electronic payments and transactions in a safe and secure manner, both nationally or cross-border, in euro and non-euro.

As regards the legislative procedure, the European Parliament adopted its first-reading position on the proposal on [open finance](#) and entered into interinstitutional negotiations.

Digital euro

The Commission tabled a [package of proposals](#) to establish a [digital euro](#) in June 2023. The [digital euro](#) would complement the euro in its physical form, but would be offered in digital form. The main motivation for the digital euro is two-fold. Firstly, the digitalisation of payment and transaction services has increased substantially. Secondly, private digital DLT-based currencies, especially stablecoins, are taking larger shares in payments and transactions. These two phenomena, combined with the EU's objective of 'strategic autonomy', are driving the digital euro as a tool to reduce external dependency on foreign providers of payment services.

Regarding the [legislative procedure](#), the Council and the European Parliament have not yet adopted their respective mandates.

Level 2 measures

Level 2 measures constitute a major component of the digital finance legislative framework and are still under construction.

Level 2 measures

The [regulatory process in financial services](#) – also called the 'Lamfalussy process' – is chiefly composed of two levels. At Level 1, the Council and Parliament adopt the 'basic act', which sets out the principles and legal framework. The basic act also empowers the Commission to adopt 'Level 2 measures'. Regulatory technical standards are adopted by means of delegated acts; implementing standards by means of implementing acts.

Markets in crypto-assets (MiCA)

Since the entry into force of MiCA in June 2023, the European Securities and Markets Authority (**ESMA**) has [launched](#) a series of consultations for packages of regulatory technical standards (**RTS**) and implementing technical standards (**ITS**). ESMA is launching the MiCA consultation packages one after another according to the respective deadlines (see Figure 1). Hence, it launched the consultations on [Package I](#), [Package II](#), and [Package III](#) in June 2023, October 2023, and March 2024, respectively.⁸

The [final report on Package I](#) was published on 25 March 2024 for rules on crypto-assets providers. It includes ITS and RTS, especially for the content of the application for authorisation (RTS), and the management, prevention and disclosure of conflicts of interest (RTS).⁹ The draft technical standards are submitted to the European Commission for adoption, which shall decide whether to adopt the technical standards within three months.

Moreover, the Commission submitted [four delegated acts](#) on 22 February 2024, currently under scrutiny by Parliament and the Council. The acts would specify the following aspects of MiCA:

- procedural rules for exercising the power to impose fines or periodic penalty payments by the European Banking Authority (**EBA**) on issuers of significant asset-referenced tokens and issuers of significant e-money tokens;
- certain criteria for classifying asset-referenced tokens and e-money tokens as significant;
- the fees charged by the **EBA** to issuers of significant asset-referenced tokens and issuers of significant e-money tokens;
- the criteria and factors to be taken into account by ESMA, the **EBA** and competent authorities in relation to their intervention powers.

Figure 1 – Markets in Crypto-assets Regulation (MiCA) timeline of Level 2 and 3 measures



Source: [European Securities and Markets Authority \(ESMA\)](#).

Consistent with Article 2(5) MiCA, ESMA conducted consultations between January and April 2024 on the '[conditions and criteria for the classification of crypto-assets as financial instruments](#)'. The aim is to establish guidelines for national competent authorities on the qualification of crypto-assets as financial instruments. ESMA is not expected to clarify the entire scope of what constitutes a financial instrument, but only products that comply with both the crypto-asset definition of MiCA and the financial instrument definition of MiFID II. MiCA sets the deadline for ESMA to issue guidelines on 30 December 2024.

Under MiCA, the **EBA** is [responsible](#) for supervising significant ARTs and EMTs (together with the national competent authority for the latter). As mandated by MiCA, the EBA has launched consultations on [recovery plans](#) to be drafted by issuers of ARTs and EMTs (closed in February 2024) and [redemption plans](#) (closure in June 2024).¹⁰

Digital Operational Resilience Act (DORA)

The ESAs published the first [set of rules](#) under DORA for ICT and third-party risk management and incident classification in January 2024. The draft technical standards include three RTSs and one ITS:

- an RTS on ICT risk management and on a simplified ICT risk management framework;
- an RTS on criteria for the classification of ICT-related incidents;
- an RTS on policy on ICT services supporting critical or important functions provided by ICT third-party service providers (TPPs);
- an IPS to establish the templates for the register of information.

On 22 February 2024, the Commission made two proposals for a delegated act in relation to DORA.

- [Determination of the amount of the oversight fees](#) to be charged by the lead overseer to critical ICT third-party service providers and the way in which those fees are to be paid.
- Specification of the criteria for [the designation of ICT third-party service providers](#) as critical for financial entities.

Positions of the co-legislators

European Parliament

The European Parliament strongly supports the digital finance strategy, and MiCA was [adopted](#) in plenary by 529 votes in favour (29 against). Parliament's positions are characterised by support for increased powers for the ESAs, both in terms of supervisory powers and setting out technical standards, and Parliament pays attention to environmental, sustainability and governance (ESG) aspects. For instance, Parliament's report on MiCA for interinstitutional negotiations enhanced the powers assigned to ESMA and the EBA, also demanding further technically specific definitions. The report introduces consideration of the environmental impact of crypto-assets and says relevant information should be provided in the information document ('white paper').

On DORA, Parliament [voted](#) in favour of the interinstitutional agreement (556 votes to 18). The ECON report for the negotiations emphasised the proportionality that should be adopted with regard to small and medium-sized enterprises (SMEs), small financial entities and other micro-enterprises. The amendments include the requirement for issuing institutions to have robust governance arrangements that include 'a clear organisational structure with well-defined, transparent and consistent lines of responsibility, effective processes to identify, manage, monitor and report the risks'.

Council

The Council has also [supported](#) the digital finance strategy, and in its [conclusion](#) on regulatory sandboxes of November 2020 it affirmed that regulatory sandboxes can offer significant opportunities for all businesses to innovate and grow, especially SMEs, including micro-enterprises and start-ups, in industry, services and other sectors. It thus encouraged the Commission to continue considering the use of experimentation clauses on a case-by-case basis when drafting and reviewing legislation, and to evaluate the use of experimentation clauses in ex-post evaluations and fitness checks based on an exchange of information with Member States

In its [statement](#) of March 2024, the Council announced three priorities and 13 measures for the future of the capital markets union, among them digitalisation of the financial system. One reason is the potential for financial institutions to create easy-to-use and secure digital interfaces that encourage citizens to access capital markets.

Authorities

In an occasional paper published in 2020, the European Central Bank (ECB) [supports](#) regulatory actions in the area of stablecoins, in any of three scenarios where stablecoins would be either an accessory to crypto-assets, a new means of payment, or a means of value storage. To ensure fair competition (a level playing field) in the industry, regulation and supervision should follow the approach of 'same business, same risks, same rules', irrespective of the underlying technology.

In its [statement](#) of March 2024, the ECB Governing Council supports the potential use of new technologies for issuance, trading and settlement, fostering tokenisation and possibly a '**European unified ledger**'. The ECB argues that this will help to promote a **digital capital markets union** and thereby strengthen the efficiency of EU financial markets while avoiding a re-fragmentation of elements that have been harmonised and integrated. The development of a unified ledger is also supported by the [Banque de France](#).

Stakeholders and academic opinions

In his [book on regulating financial innovation](#) published in 2023, Professor Ruof at Hamburg University argues that there has been a significant shift from the negative view of financial innovation that dominated in the aftermath of the 2008 financial crisis; today, the primary goal of regulators is to promote innovation. While these innovation-friendly regulatory initiatives are relatively nascent, they have kicked off a competition among regulators worldwide on attracting the best financial technology innovations and firms. However, the author argues that the task of the regulator is complex, because information about financial innovation is dispersed and disaggregated among countless individuals. In that sense, the regulatory sandbox represents a move towards a more principles-based approach, taking more into account the individual risk profile of a firm as opposed to merely applying the existing regulatory framework. This feature makes it considerably more attractive than the innovation hub for financial technology firms. As a result, innovative firms might be attracted to the regulatory sphere, while at the same time experimentation in the sandbox produces information for the regulator. The opportunity provided by sandboxes is highlighted and welcomed by several [legal firms](#).

Sandboxes have also become a regulatory instrument of notable interest for academics. In their 2020 [study](#), researchers from the Bank for International Settlements argue that the potentially disruptive growth of firms offering innovative products and services poses new challenges for financial stability and consumer protection. 'Regulatory sandboxes' come as a response to such a challenge, by supporting innovation in the financial sector while monitoring emerging risks. According to the authors, the key objective of sandboxes is to facilitate start-ups' access to capital. Despite this challenge, financial technologies will be beneficial to the sector, especially thanks to efficiency gains, a wider choice of products for consumers, and enhanced financial inclusion.

In an [academic analysis](#) published in 2021, Professor McCarthy argues that, although there is continued ambiguity as to the characteristics of sandboxes, broader regulatory support, such as innovation hubs, can be vital. However, the author adds that this should not undermine the need for improved clarity and transparency regarding sandboxes' operations.

On the digital euro, in an [opinion piece](#) published by Bruegel in June 2023, the authors argue that cross-border digital currency payments could change the financial system. The authors note that, while there are now more than 110 countries exploring the creation of a digital currency, the euro and the US dollar are still at early stages compared with other countries and would need to accelerate. Moreover, the EU needs to further address the cross-border issues relating to the international role of the euro.

On DORA, professional services firm [Deloitte](#) stresses that the adoption of the RTS is an important part of the regulation, and firms will not know the full spectrum of requirements until they are adopted. According to Deloitte, preparing for the initial implementation of the new rules has taken

more time and resources than many firms anticipated. Consulting company [EY](#) confirms DORA will require financial services to embed digital resilience at all levels of their operations, and they urgently need to prepare a roadmap to achieve compliance. According to EY, DORA requires a complete view of the ICT landscape that supports critical business functions, as well as a thorough approach to 'business continuity, incident management and third-party risk'. Moreover, firms can expect a higher degree of supervision from ESAs and stronger controls, with obligations.

In an [academic analysis](#), Professor Pavlidis argues that the exponential expansion of the cryptocurrency market in the last 10 years made action necessary; the EU digital finance strategy is filling the regulatory gaps and modernising the EU economy. The author argues that the philosophy behind the strategy, especially MiCA and DORA, 'remains innovation-friendly and will lead to additional and better financial products for consumers, improving financial inclusion and financing of businesses'.

In a more recent [empirical study](#) published in 2021, economists from the Bank for International Settlements and Paris Nanterre University looked into the effects of the UK sandbox. They find evidence that a firm's entry into the sandbox is associated with an increase of about 50 % in the probability of raising capital. Furthermore, the sandbox is particularly effective at reducing informational issues and regulatory costs. The positive effect of sandbox entry on capital raised is particularly pronounced for smaller and younger firms, i.e. firms that are usually considered more opaque and hence subject to more severe informational issues.

Finally, in a [blueprint](#) for the future of the monetary system published in June 2023, the Bank for International Settlements argues that tokenisation of money and assets has great potential, but that initiatives to date have taken place in silos without access to central bank money and the underpinning trust it provides. A unified ledger would establish a new financial architecture that could capture the full benefits of tokenisation by combining central bank money, tokenised deposits and tokenised assets on a programmable platform. It would improve existing processes and enable arrangements that are currently not practicable. Multiple ledgers – each with a specific use case – might coexist, but would be interlinked by application programming interfaces. This would ensure interoperability while also promoting financial inclusion and a level playing field.

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ENDNOTES

- ¹ The [debate](#) about regulating stablecoins is an example, but not the only one. The regulator first needs to determine whether the existing regulation of financial intermediaries such as banks and insurance firms – essential for financial stability – covers the new financial technologies and instruments, or whether new rules are required.
- ² See Hallak I., [Capital Markets Union: overview and state of play](#) (2024), for a discussion of the impact of capital market 'defragmentation', i.e. integration, on the effectiveness of the single market.
- ³ [Regulation \(EU\) 2022/858](#) on a pilot regime for market infrastructure based on DLT, Article 4(1), point (15) was amended as follows: "financial instrument" means those instruments specified in Section C of Annex I, including such instruments issued by means of distributed ledger technology'. The regulation on a DLT pilot regime is discussed further in the following sections.
- ⁴ The Bank for International Settlements [defines](#) distributed ledger technology (DLT) as the 'protocols and supporting infrastructure that allow computers in different locations to propose and validate transactions and update records in a synchronised way across a network'. MiCA [Article 3\(1\)-\(2\)](#) defines DLT as 'a technology that enables the operation and use of distributed ledgers', i.e. 'an information repository that keeps records of transactions and that is shared across, and synchronised between, a set of DLT network nodes using a consensus mechanism'.
- ⁵ In a [letter](#) to the institutions in April 2024, ESMA reports that four official applications have been submitted. Around eight other potential applications may be submitted during the course of this year. ESMA also identifies challenges that could explain the low number of applications.
- ⁶ Note that MiCA Article 143 provides Member States with options to implement transitional measures, including:
 - a 'grand-fathering' clause allowing entities providing crypto-asset services in accordance with national applicable laws before 30 December 2024 to continue to do so until 1 July 2026 or until they are granted or refused a MiCA authorisation (MiCA Article 143(3));
 - a simplified authorisation procedure for entities that were already authorised under national applicable law on 30 December 2024 to provide crypto-asset services (MiCA Article 143 (6)).
- ⁷ See also the EPRS [legislative train](#).
- ⁸ All ESMA's consultations on digital finance and innovation are available on their [search site](#).
- ⁹ MiCA Articles 62(5) and 72(5), respectively. Other acts include:
 - Content of notification from selected entities to national competent authorities (MiCA Article 60(13), RTS).
 - Forms and templates for notification from entities to national competent authorities (MiCA Article 60(14), ITS).
 - Forms and templates authorisation application (MiCA Article 62(6), ITS).
 - Complaint handling procedure (MiCA Article 71(5), RTS).
 - Intended acquisition information requirements (MiCA Article 84(4), RTS).
- ¹⁰ The EBA's [search site](#) contains the latest documents relating to consultations on MiCA (called MiCAR by the EBA).

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